Tuning report: tropical / MinCoinChange

2026-10-22T19:17:04

1 Decision

Approving freezes `fishr.lambda` = 10 into every cell of the grid (frozen hash `c61c89bd8676`). The experiment YAML has `fishr.lambda` = 1000 for this campaign. `frozen.yaml` is currently unapproved with {'fishr.lambda': 10.0}.

Selection rule. The candidate with the lowest mean `id_val_loss` wins, averaged over 6 cell(s) and 2 tuning seed(s) each (global seed 1000, disjoint from the grid's 0), across 4 candidates. A candidate with a missing or diverged run is not eligible.

Verdict: REVIEW

- **review**: the winner is not distinguishable from fishr.lambda=1000: the paired gap's 95% CI is [-0.00042, 0.00208]
- **review**: fishr.lambda=10 is the smallest value searched: the optimum may lie beyond it, so widen the space
- **review**: 9 of 12 winner runs reached their best id_val_loss at the last logged epoch: the search budget may be too short to rank candidates
- **review**: 9 of 12 winner runs picked their last epoch (best_epoch = 19): still improving when the search budget ran out, so candidates were ranked before converging
- **info**: the search ran with optim.epochs=20; the grid runs with optim.epochs=100
- **info**: devices: cuda:0

block: do not approve; fix the search and re-run it. **review**: read the sections below before approving. **info**: context only.

2 Search setup

- `fishr.lambda` ∈ {10, 100, 1000, 1e+04}
- Cells trained (`tune.only` {'objective': 'fishr'}):
  - `model_type=vanilla head=seq2seq objective=fishr`
  - `model_type=vanilla head=pointer objective=fishr`
  - `model_type=adaptive head=seq2seq objective=fishr`
  - `model_type=adaptive head=pointer objective=fishr`
  - `model_type=tropical head=seq2seq objective=fishr`
  - `model_type=tropical head=pointer objective=fishr`
- Search-only overrides (never frozen): {'optim': {'epochs': 20}}. The chosen value was ranked under this budget, not the grid's.

3 Candidates

Table 1 ranks the candidates on `id_val_loss` (lower is better); Figure 1 plots them. Candidates share seeds, so Table 2 compares each with the winner run by run (same cell, same seed).

Table 1: Every candidate: rank, runs stored/expected, diverged, mean with 95% bootstrap CI, SD and range over all its runs.

#	`fishr.lambda`	runs	div.	mean	95% CI	SD	range	
1	10	12/12	0	0.588	[0.582, 0.596]	0.0123	[0.576, 0.618]	**selected**
4	100	12/12	0	0.5889	[0.583, 0.596]	0.012	[0.574, 0.619]	
2	1000	12/12	0	0.5887	[0.583, 0.596]	0.0119	[0.573, 0.618]	
3	1e+04	12/12	0	0.5889	[0.583, 0.596]	0.0116	[0.575, 0.618]	

Table 2: Each candidate minus the winner on the runs they share, signed so a positive gap is worse than the winner. Worse/better/equal count the runs; p is a two-sided sign test.

candidate	n	gap	95% CI	worse	better	equal	p
`fishr.lambda=100`	12	8.227×10^{-4}	$[-1.87 \times 10^{-4}, 0.00214]$	5	4	3	1.0000
`fishr.lambda=1000`	12	6.806×10^{-4}	$[-4.20 \times 10^{-4}, 0.00208]$	4	5	3	1.0000
`fishr.lambda=10000`	12	8.216×10^{-4}	$[-1.55 \times 10^{-4}, 0.00209]$	4	5	3	1.0000

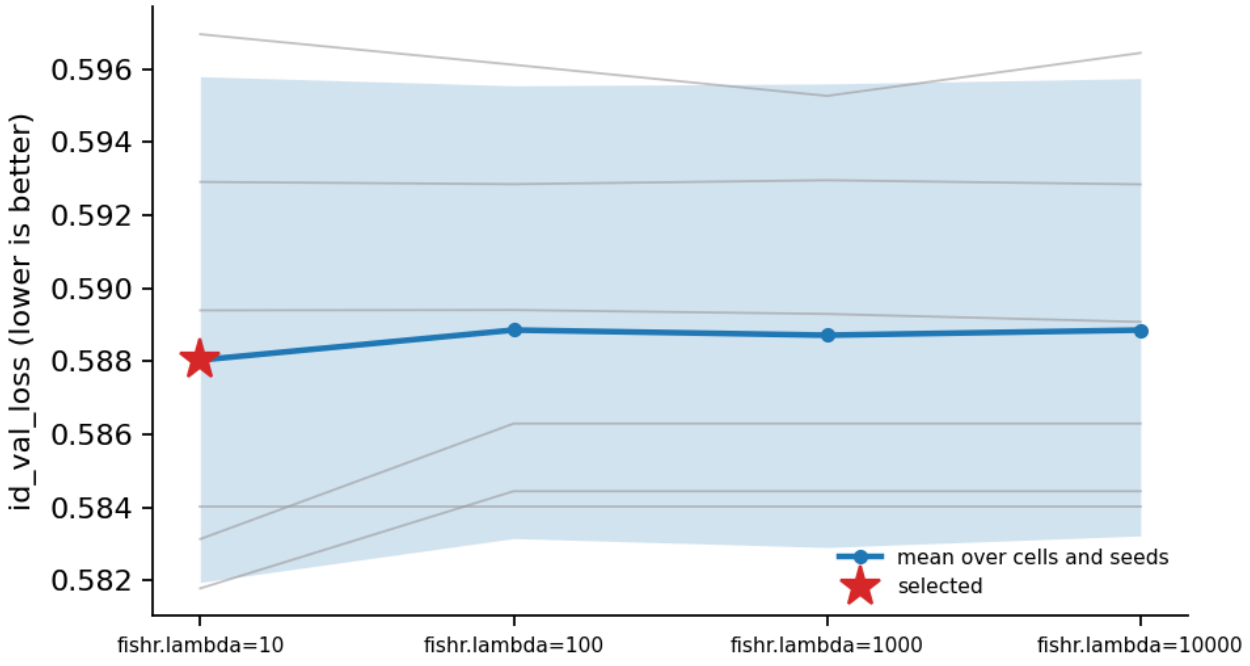


Figure 1: `id_val_loss` per candidate: the mean over every run with a 95% bootstrap CI (blue), each cell's seed-mean (grey), the selected candidate (red star).

4 Per cell

One value is frozen for every cell, chosen by the mean over cells. Figure 2 shows whether the winner is best in each cell or is carried by a few of them.

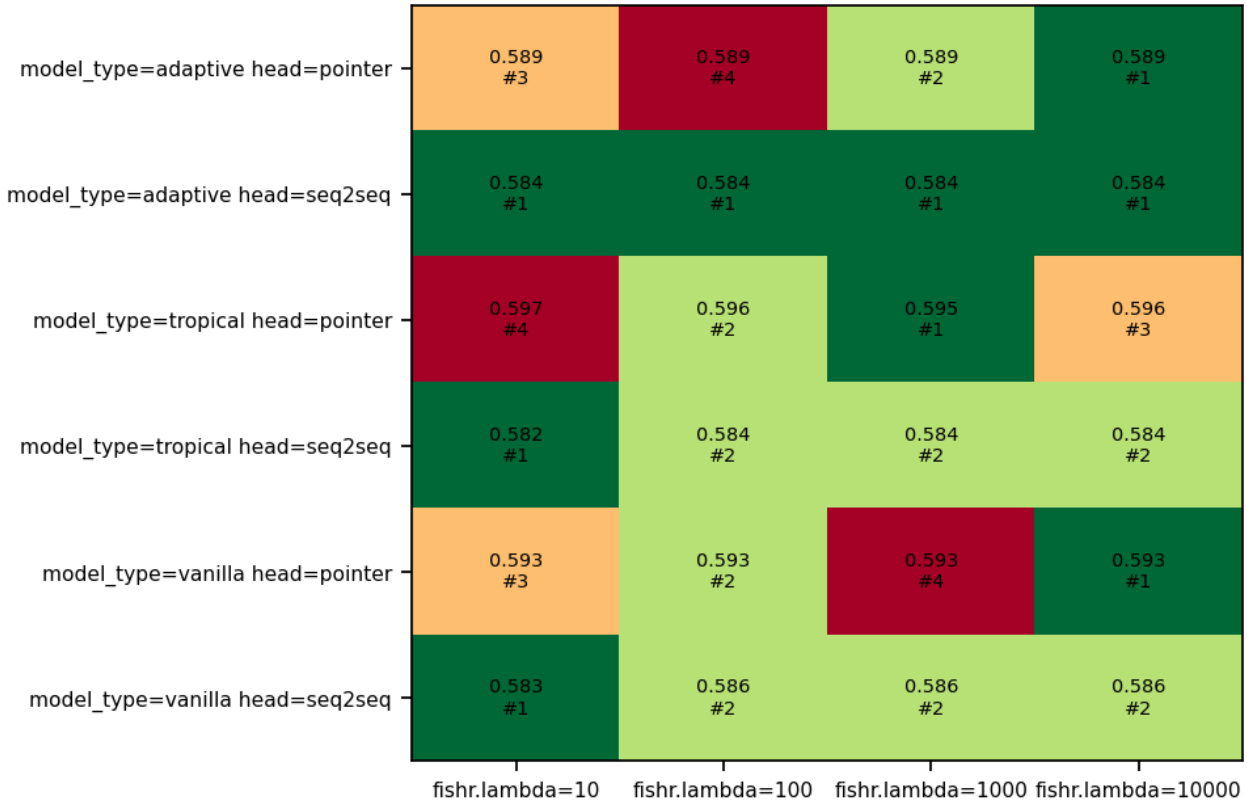


Figure 2: Each cell's seed-mean `id_val_loss` per candidate, with its rank within the cell (#1 is best; green is better).

Table 3: The best candidate in each cell.

cell	best
`model_type=adaptive head=pointer`	fishr.lambda=10000
`model_type=adaptive head=seq2seq`	fishr.lambda=10, fishr.lambda=100, fishr.lambda=1000, fishr.lambda=10000
`model_type=tropical head=pointer`	fishr.lambda=1000
`model_type=tropical head=seq2seq`	fishr.lambda=10
`model_type=vanilla head=pointer`	fishr.lambda=10000
`model_type=vanilla head=seq2seq`	fishr.lambda=10

5 Other training-distribution metrics

Not used for selection; shown so a winner that trades them away is visible.

Table 4: Mean of every other in-distribution metric per candidate.

candidate	`id_f1`	`id_exact`	`fishr_penalty_final`	`best_epoch`
fishr.lambda=10	0.399	0.172	0.0148	15.8

Table 4: Mean of every other in-distribution metric per candidate.

candidate	id_f1	id_exact	fishr_penalty_final	best_epoch
fishr.lambda=100	0.399	0.171	0.00811	12.5
fishr.lambda=1000	0.399	0.17	0.0055	12.5
fishr.lambda=10000	0.399	0.171	0.00605	12.5

6 Training curves

The mean over every cell and seed of each candidate; the selected candidate in bold red.

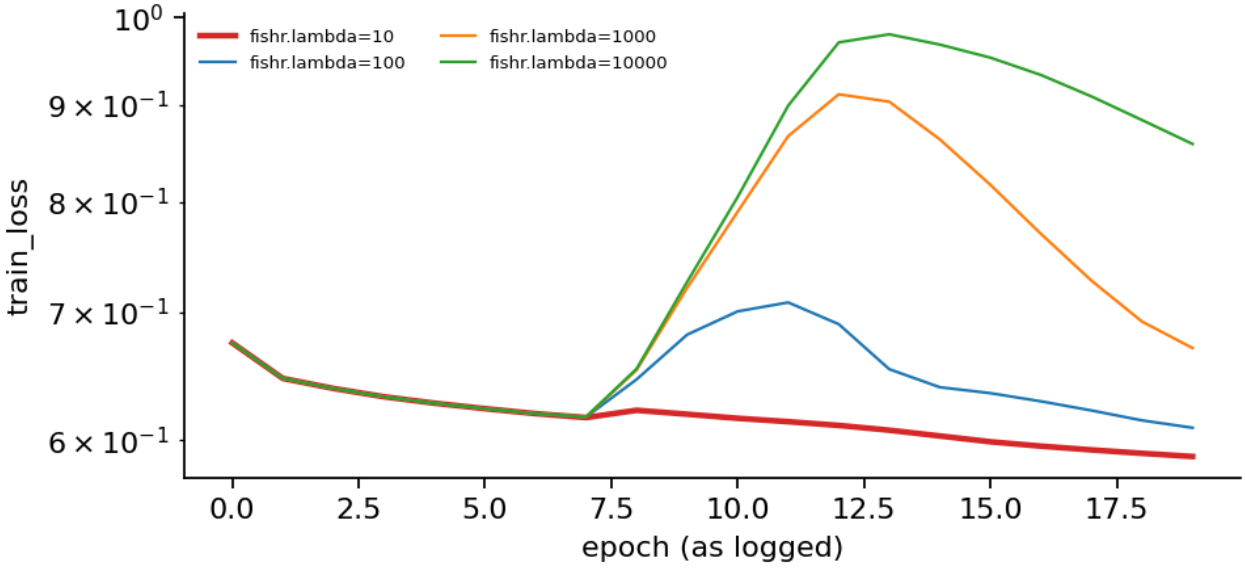


Figure 3: `train_loss` per candidate.

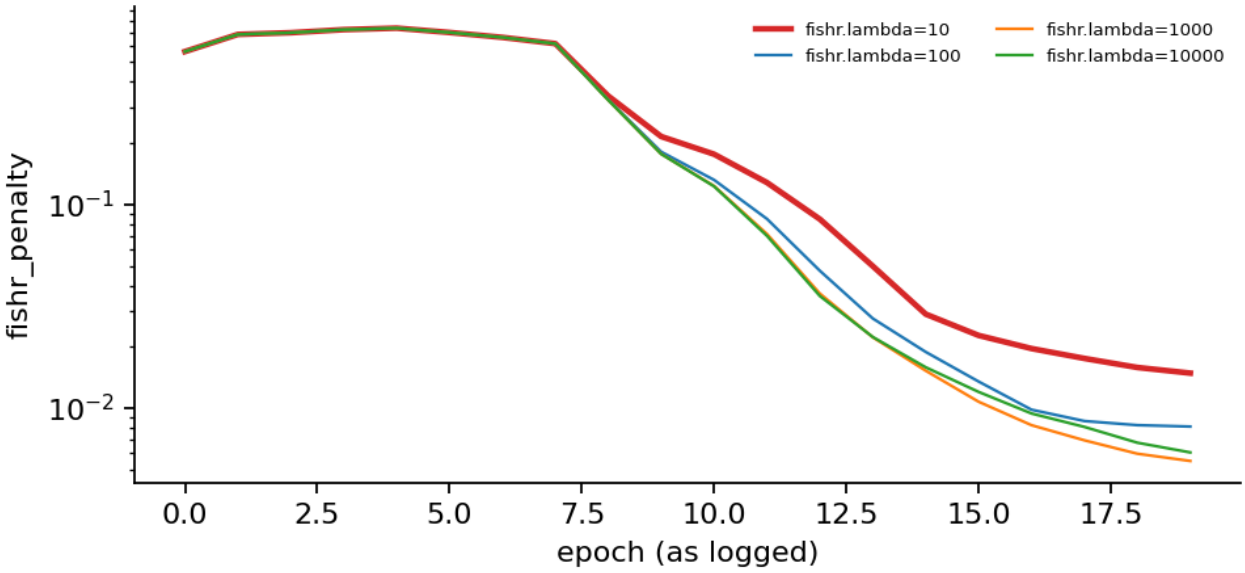


Figure 4: `fishr_penalty` per candidate.

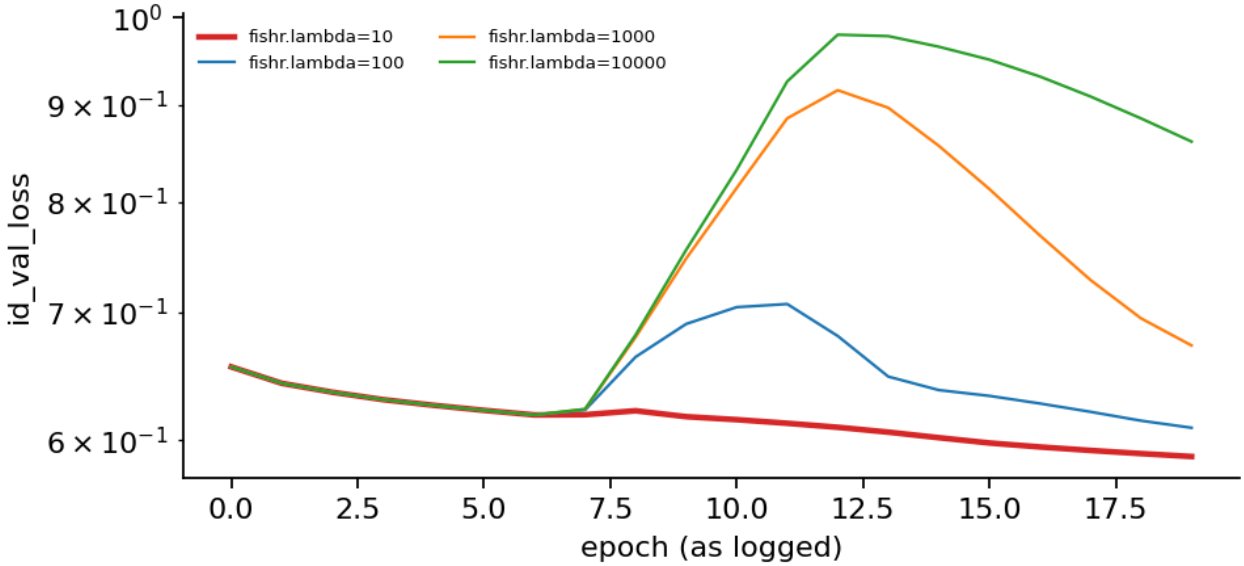


Figure 5: `id_val_loss` per candidate.

7 Provenance

Table 5: Each candidate's runs: base hash, runs, total wall time, commits (+ marks a dirty tree), torch.

candidate	hash	runs	wall (s)	commit	torch
fishr.lambda=10	e8f341121ff0	12	3.63e+03	e5ccce8034	2.8.0+cu128
fishr.lambda=100	5ac1b063fb5f	12	3.64e+03	e5ccce8034	2.8.0+cu128
fishr.lambda=1000	3c27b131aaf8	12	3.63e+03	e5ccce8034	2.8.0+cu128
fishr.lambda=10000	87eaff28224f	12	3.64e+03	e5ccce8034	2.8.0+cu128

Store: `campaigns/tropical/MinCoinChange/tune/results`.
Rebuild: `expkit tune-report --config configs/tropical.yaml --profile MinCoinChange`
Approve: `expkit approve --config configs/tropical.yaml --profile MinCoinChange`

The base the grid would run with.

```
task:
  name: MinCoinChangeDataset
  value_range:
  - 1
  - 10
  target_range:
  - 10
  - 20
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  lambda: 10.0
  ema: 0.95
```

```
anneal_iters: 1500
reset_optimizer: true
```

A Sealed: OOD and test metrics

Read this only after deciding. These were recorded by the tuning runs but never consulted by the selection. Choosing a value because it looks better here would tune on the test distribution.

Table 6: Mean of every OOD or test metric per candidate.

candidate	ood_len_f1	ood_len_exact	ood_val_f1	ood_val_exact	ood_noise_f1	ood_noise_exact	ood_len_f1_L12	ood_len_exact_L12	ood_len_f1_L16	ood_len_exact_L16	ood_len_f1_L64	ood_len_exact_L64
fishr.lambda=10	0.21	0.0215	0.132	0.293	0.254	0.12	0.37	0.0414	0.323	0.0227	0.124	4.58×10^{-4}
fishr.lambda=100	0.219	0.0241	0.131	0.312	0.254	0.119	0.372	0.0446	0.33	0.0266	0.141	9.17×10^{-4}
fishr.lambda=1000	0.219	0.0235	0.137	0.307	0.254	0.119	0.372	0.0436	0.33	0.0261	0.141	9.17×10^{-4}
fishr.lambda=10000	0.218	0.0239	0.136	0.305	0.254	0.119	0.371	0.0441	0.329	0.0266	0.14	9.17×10^{-4}